



# ELE1204: Digital Electronics and Logic Design

## Lecture 09 (BSCS)

### *Finite State Machines (FSM)*

Ms. Olivia Nakayima

(MSc, BSc. CCNA, JNCIA, JNCIS)

Department of Computing & Technology

Faculty of Engineering, Design & Technology

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# Lecture Objectives and Learning outcomes

The Objectives of this lecture are :

- ☐ To understand what FSM is and its role in sequential circuits
- ☐ To understand the different types of FSM and their implementation

NOTE:: SMART = Specific, Measurable, Achievable, Relevant, Timebound

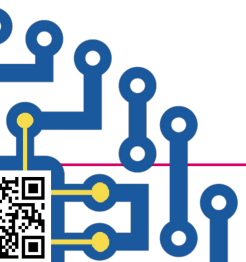
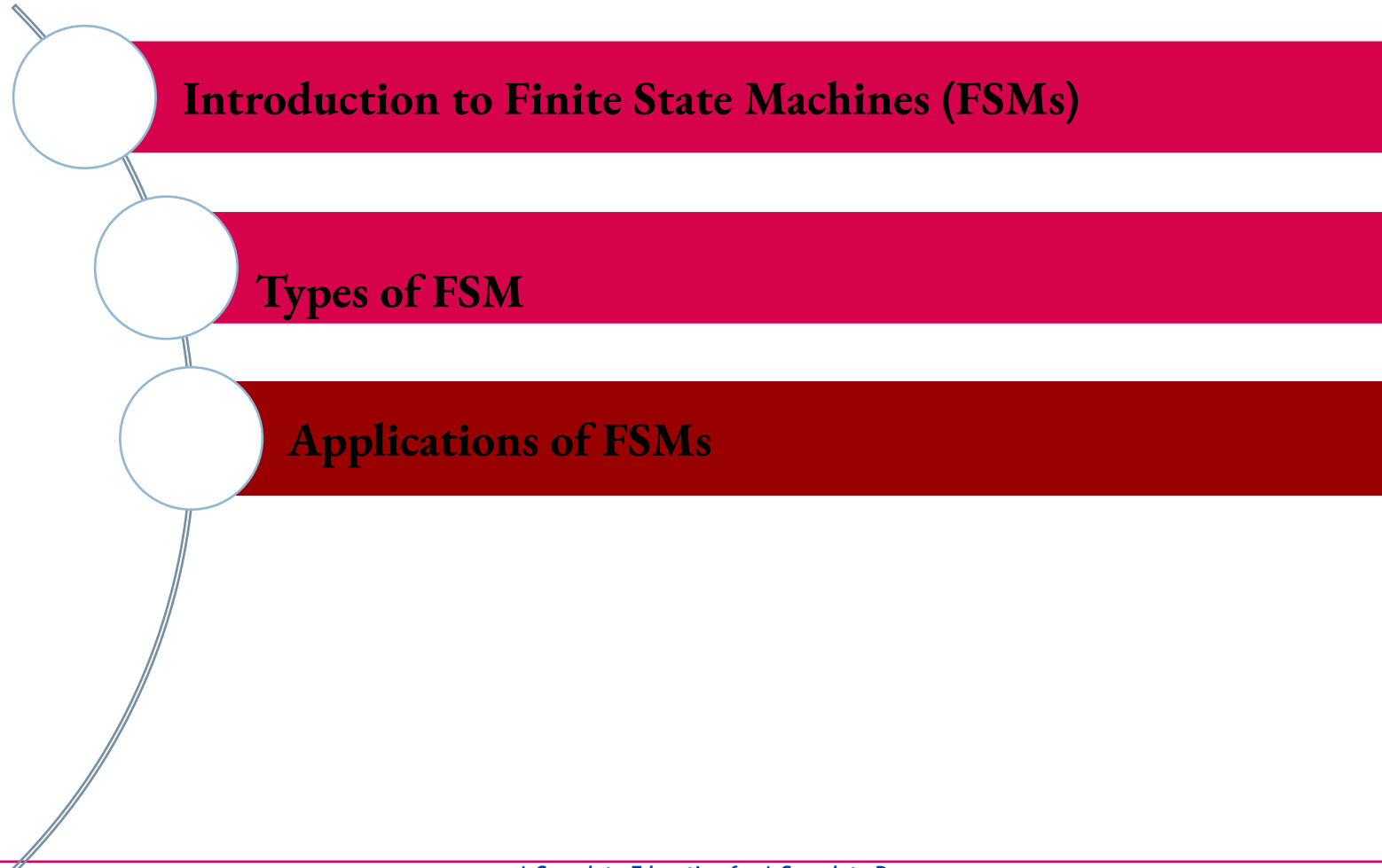
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# Lecture Overview





## Introduction to Finite State Machines (FSMs)

A **Finite State Machine (FSM)** is a **sequential logic model** used to design digital circuits that transition between states based on inputs and **clock signals**. FSMs are widely used in **embedded systems, digital logic design, robotics, and control systems**. Why finite (it has a finite number of internal states):

<https://www.youtube.com/watch?v=kb-Ww8HaHuE>

### Types of Finite State Machines

FSMs are classified into **two main types**:

| Type          | Output depends on | characteristics                                       |
|---------------|-------------------|-------------------------------------------------------|
| Mealy Machine | Input and state   | Faster response, output can change within a state     |
| Moore machine | Only states       | More stable, output changes only on state transitions |



# Mealy Machine

In a **Mealy Machine**, the **output depends on both the current state and the current input**. The output can **change instantly** when inputs change.

## Example: Traffic Light Controller (Mealy)

- If a pedestrian presses a button, the light transitions immediately.

## Mealy Machine Representation

- **State Transition Diagram:** Uses circles (states) and labeled arrows (transitions).
- **State Table:** Lists current states, inputs, next states, and outputs.

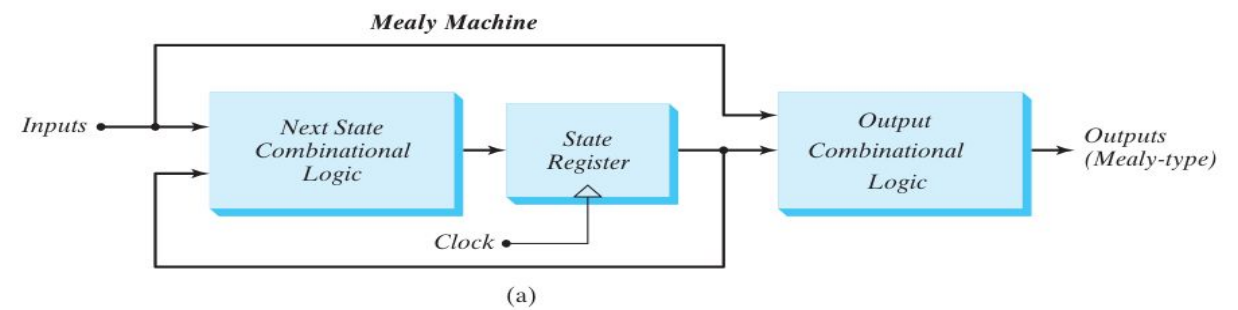
## Advantages of Mealy Machines:

Respond **immediately** to input changes.

Require **fewer states** than Moore Machines.

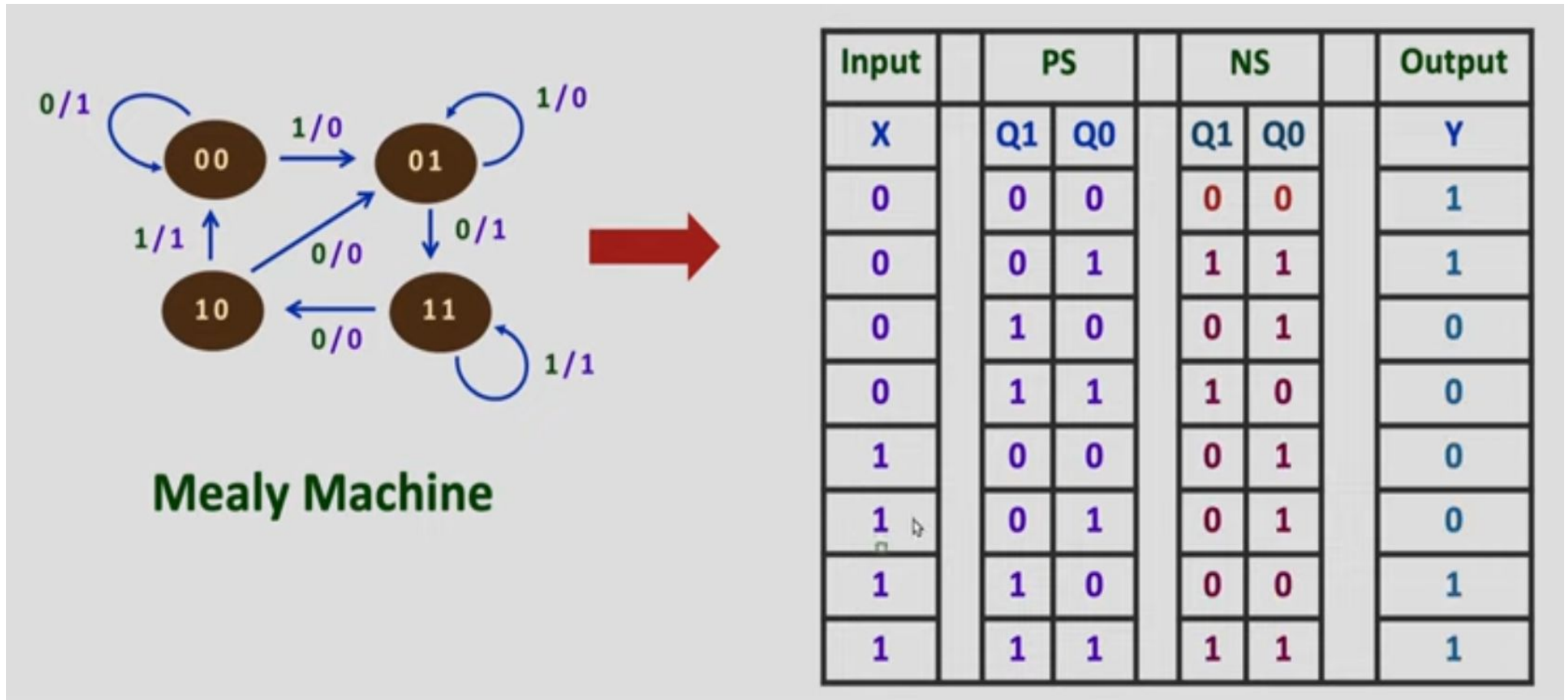
## Disadvantages of Mealy Machines:

Output changes within a state, making it sensitive to glitches.



## Mealy Machine - Example

**NB.**  
Input/Output  
Left/right



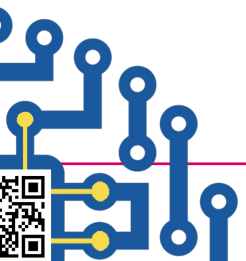
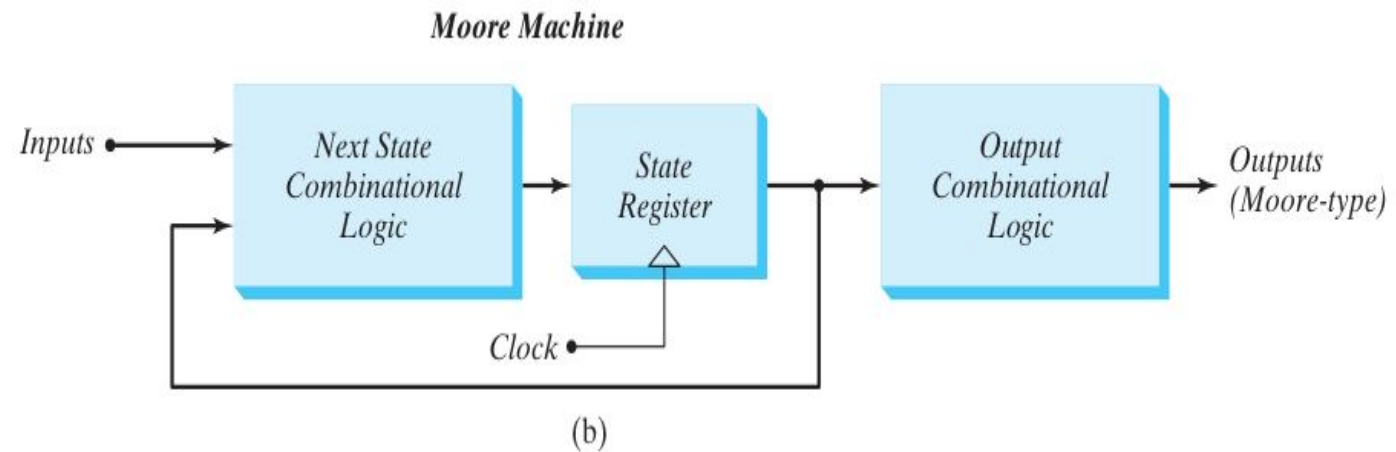
# Moore Machine

In a **Moore Machine**, the **output depends only on the current state**, NOT directly on the input. The output **remains stable** within a state and only changes **on state transition**.

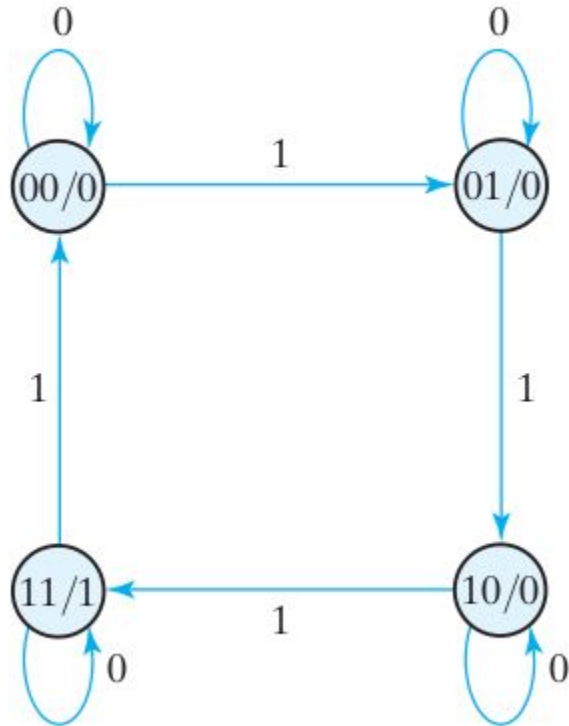
## Example: Elevator Controller (Moore)

- The floor number display updates only after reaching a floor, NOT while moving.

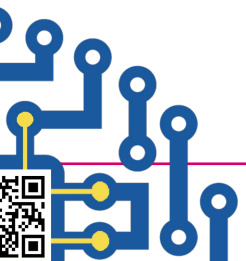
## Moore Machine Representation



## Moore Machine - Example



| Present State |     | Input<br>$x$ | Next State |     | Output<br>$y$ |
|---------------|-----|--------------|------------|-----|---------------|
| $A$           | $B$ |              | $A$        | $B$ |               |
| 0             | 0   | 0            | 0          | 0   | 0             |
| 0             | 0   | 1            | 0          | 1   | 0             |
| 0             | 1   | 0            | 0          | 1   | 0             |
| 0             | 1   | 1            | 1          | 0   | 0             |
| 1             | 0   | 0            | 1          | 0   | 0             |
| 1             | 0   | 1            | 1          | 1   | 0             |
| 1             | 1   | 0            | 1          | 1   | 1             |
| 1             | 1   | 1            | 0          | 0   | 1             |







# Moore Machine

## Advantages of Moore Machines:

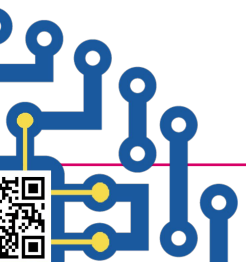
More **stable**, since output changes only on state transitions.

Less sensitive to **glitches** and **input noise**.

## Disadvantages of Moore Machines:

Can require **more states** than Mealy Machines for the same function.

| Feature              | Mealy                | Moore                  |
|----------------------|----------------------|------------------------|
| Output depends on    | Input and state      | Only state             |
| Output change timing | Immediate            | After state transition |
| Number of states     | Fewer                | More                   |
| Glitch sensitivity   | More sensitive       | Less sensitive         |
| Use case example     | Traffic light system | Elevator controller    |





## Applications of FSMs

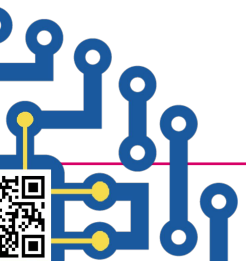
**Traffic Light Controllers** – Control traffic signals based on sensors and timers.

**Elevators** – Moves between floors based on user input.

**Vending Machines** – Dispenses items based on payment input.

**Robotics** – Controls robotic movement and decision-making.

**Communication Protocols** – Handles data transmission sequences (e.g., UART, SPI, I2C).





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## Department of Computing & Technology FACULTY OF ENGINEERING, DESIGN AND TECHNOLOGY

Tel: +256 (0) 312 350 863 | WhatsApp: +256 (0) 708 114 300

 @ucuc Computeng  @ucu\_ComputEng

 <https://cse.ucu.ac.ug/> Email: [dct-info@ucu.ac.ug](mailto:dct-info@ucu.ac.ug)

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